

EXPERIMENT 24: Prolog Program to Suggest Dieting System Based on Disease

Aim

To write and execute a Prolog program that suggests a suitable diet based on the disease specified by the user using facts and logical rules.

Algorithm

1. Start the Prolog program.
2. Define facts representing diseases and recommended food items.
3. Define rules connecting diseases with suitable diet recommendations.
4. Accept the disease as input through a query.
5. Match the disease with the corresponding rule.
6. Display the recommended diet.
7. Stop.

Program Code:

```
% Disease-based diet suggestion system
```

```
diet(diabetes, 'Low sugar and high fiber diet').
```

```
diet(hypertension, 'Low salt and low fat diet').
```

```
diet(obesity, 'Low calorie and high protein diet').
```

```
diet(anemia, 'Iron rich foods such as spinach and beans').
```

```
diet(heart_disease, 'Low fat and low cholesterol diet').
```

```
% Suggest diet
```

```
suggest_diet(Disease) :-
```

```
    diet(Disease, Advice),
```

```
    write('Recommended diet: '),
```

```
    write(Advice),
```

```
    nl.
```

suggest_diet(Disease) :-

\+ diet(Disease, _),

write('No diet information available. '), nl.

OutPut:

```
\-- --/ --
| ?- ndparent(john, alex).
uncaught exception: error(existence_error(procedure,ndparent/2),top_level/0)
| ?- consult('C:/Users/Hp/Documents/ex-24.pl').
compiling C:/Users/Hp/Documents/ex-24.pl for byte code...
C:/Users/Hp/Documents/ex-24.pl compiled, 17 lines read - 1671 bytes written, 17 ms

yes
| ?- suggest_diet(diabetes).
Recommended diet: Low sugar and high fiber diet

true ?

(31 ms) yes
| ?- suggest_diet(obesity).
Recommended diet: Low calorie and high protein diet

true ?

yes
\-- --/ --
```

Result

The Prolog-based diet suggestion system was successfully implemented. Based on the specified disease, the program correctly provided the corresponding diet recommendation using Prolog rules and queries.